Emiliano Méndez Jiménez

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Summary

Robotics undergraduate who deploys and extends recent robot-learning methods on real hardware (Unitree G1 humanoid, Lite3 quadruped). *Interests:* humanoid whole-body control · sim-to-real RL · VLA models · semantic navigation.

Education

[2024 – 2028] B.S. Robotics and Digital Systems, Tecnológico de Monterrey (ITESM), Mexico GPA 92/100

Experience

Humanoid Team Captain & Founder, VantTec, Tec de Monterrey [Monterrey, MX] Aug 2025 – pres

- Founded and lead the campus's first humanoid team (18 members, 6 subteams) around a Unitree G1 EDU.
- Deployed BFM-Zero, a behavior foundation model, onboard the G1 (Jetson Orin NX), voice-controlled by an LLM agent.
- Trained a BeyondMimic-style tracking policy (mjlab) for 8 boxing strikes and demoed it live by voice on the real robot at Venture Café's AI Summit (Oct 2026).
- Built XR whole-body teleoperation (Quest 3, dexterous hands, 100 Hz recording) for VLA data, a video-to-robot retargeting pipeline (GEM-X, MuJoCo sim-to-sim), and an EEG motor-imagery-to-grasp bridge with Cinvestav BCI researchers.

Research Intern, Astrocor [Monterrey, MX] Sep 2026 – pres

- Deployed OneMap (KTH, ICRA 2025), zero-shot object search, on a DEEP Robotics Lite3 quadruped (ZED X, Jetson AGX Orin), running FP16 CLIP, YOLOv7 and MobileSAM onboard in closed loop.
- Integrated OneMap and NVIDIA's ReMEmbR (VLM+LLM memory) into Astrocor's photorealistic simulator for a blind benchmark with identical scenes, start poses and metrics, as baselines for Astrocor's own method.

Computer Vision Engineer, ADA Labs [Remote] Jun 2026 – Sep 2026

- As sole CV engineer, shipped tomato counting (YOLOv8 + ByteTrack) on a greenhouse AGV at 96% accuracy (~1,200 tomatoes per row, 20+ rows), fusing AGV pose and camera extrinsics into per-tomato 3D profiles.
- Built the Grounding DINO + SAM2/3 auto-annotation pipeline behind the training data, and an AWS GPU inference engine for the AGV digital twin that helped win 2026 Latitud Fellowship funding.

Robotics Software Engineering Intern, AIRDX Technologies [Monterrey, MX] Dec 2025 – Mar 2026

- Led software for an aircraft-inspection drone: GPS-denied indoor flight with Point-LIO odometry fed to a CubePilot via MAVROS on a Jetson, plus a YOLO lightning-strike damage detector.

UAV Team Captain, VantTec, Tec de Monterrey [Monterrey, MX] May 2025 – Dec 2025

- Led a 15-member drone team (5th place, Torneo Mexicano de Robótica); flew Monterrey's first autonomous drone pizza delivery.

Presentations and Publications

Méndez Jiménez, E. (2026). *Physical AI, Live: Humanoid and Quadruped Demos*. Live demo, Venture Café AI Summit.

Méndez Jiménez, E. et al. (2026). *Bridging the Educational Gap in Robotics: A Hands-on Tutorial Framework for Reinforcement Learning*. Poster, IEEE/SICE SII 2026. Based on our 6-h tutorial at LARS 2025 (2 editions).

Awards and Outreach

[SEP 2026] 1st place, Carrier "Keep It Cool" Hackathon (SentinelBox: neural-network HVAC health monitoring)
[2025 – pres] Volunteer instructor, SMALC: middle-school robotics course, from Python to autonomous Tello drones

Skills

Robot learning	Isaac Lab/Sim, MuJoCo, mjlab, PPO, sim-to-real, motion retargeting, BFM-Zero, GR00T WBC
Perception	PyTorch, YOLO, SAM2/3, Grounding DINO, CLIP, ByteTrack, OpenCV, LiDAR-inertial odometry
Systems	Python, C++, ROS 2, DDS, Docker, Jetson Orin, ONNX/TensorRT, AWS, STM32, FPGA (VHDL)
Languages	Spanish (native), English (C1), German (B1)

Last updated: November 9, 2026